Supplementary Material

The Ranking Transfers, the Threshold Does Not: Early At-Risk Prediction in Unseen Courses

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All numbers in this document are generated by the same script as the article (`src/08_paper_assets.py`) from the outputs of the public repository. Unless stated otherwise: cut-off 25% of the presentation, students still enrolled at the cut-off, protocol (B) is the strict leave-one-module-out, means are over the seven modules.

S1 Remedies per module

Table S1 details, module by module, the comparison summarized in Table V of the article (random sample of $k = 100$ labelled students).

Table S1: Remedies per module (25% cut-off, random sample of $k = 100$ labelled students of the target module for Platt and the intercept-only update, mean of 50 draws, standard deviation across draws in parentheses; all rules are evaluated on the remaining students). Top: flag-rate drift Δ^{ref} against the absolute-feature model with the module seen. Bottom: precision. Rules as in Table V of the article.

Module	Raw	Platt	Intercept	Top-q	Top-q (EM)	Relative
Flag-rate drift Δ^{ref}						
AAA	+0.10	+0.07 (0.05)	+0.07 (0.05)	+0.07	−0.05	+0.02
BBB	+0.19	+0.06 (0.04)	+0.06 (0.04)	+0.05	+0.37	+0.03
CCC	−0.03	+0.03 (0.03)	+0.03 (0.03)	−0.02	−0.05	+0.02
DDD	−0.05	+0.02 (0.03)	+0.01 (0.03)	−0.04	−0.11	−0.05
EEE	+0.05	+0.10 (0.04)	+0.08 (0.04)	+0.15	−0.01	+0.03
FFF	−0.03	+0.07 (0.04)	+0.07 (0.04)	+0.05	−0.16	+0.01
GGG	+0.51	−0.05 (0.04)	−0.03 (0.04)	−0.03	+0.56	−0.14
Mean $\lvert\Delta^{\text{ref}}\rvert$	0.14	0.06	0.05	0.06	0.19	0.04
Mean $\lvert$flagged − risk$\rvert$	0.17	0.06	0.06	0.07	0.22	0.06
Precision						
AAA	0.38	0.39	0.38	0.38	0.42	0.45
BBB	0.54	0.60	0.60	0.61	0.47	0.67
CCC	0.78	0.74	0.74	0.77	0.69	0.72
DDD	0.79	0.75	0.76	0.78	0.81	0.81
EEE	0.70	0.66	0.67	0.60	0.74	0.72
FFF	0.76	0.69	0.70	0.70	0.87	0.76
GGG	0.38	0.55	0.54	0.56	0.37	0.64
Mean	0.62	0.62	0.63	0.63	0.62	0.68

S2 Cohort size

Table S2 gives the performance of the rank model when ranks are computed in random sub-cohorts of the target module, over 20 random partitions. From about 50 students, ranks are as reliable as with the whole presentation; below 20, the gain degrades sharply.

S3 Average precision

Average precision (area under the precision–recall curve) leads to the same conclusions as the AUC (Table S3), including for the rarer "Withdrawn only" target (prevalence 0.137).

Table S2: Ranks computed in random sub-cohorts of n students of the target module (25% cut-off, protocol (B), means over modules and over 20 random partitions; interval: 2.5–97.5% of the module mean across partitions). Shift: absolute change of the share flagged relative to ranks computed on the whole presentation.

n	AUC (B)	95% interval	Shift	|flagged − risk|	Precision	F1
10	0.778	[0.773, 0.781]	0.042	0.082	0.677	0.617
20	0.794	[0.791, 0.797]	0.016	0.065	0.675	0.641
50	0.802	[0.800, 0.804]	0.006	0.059	0.676	0.651
100	0.805	[0.804, 0.806]	0.004	0.058	0.678	0.655
200	0.806	[0.805, 0.807]	0.003	0.058	0.678	0.656
whole presentation	0.808	—	0.000	0.057	0.680	0.658

Table S3: Average precision (area under the precision–recall curve) at the 25% cut-off, means over modules.

Target	Model	Features	Prevalence	AP (A)	AP (B)
W only	forest	absolute	0.137	0.231	0.212
W only	forest	relative	0.137	0.250	0.230
W or F	boosting	absolute	0.387	0.748	0.728
W or F	boosting	relative	0.387	0.770	0.762
W or F	forest	absolute	0.387	0.736	0.720
W or F	forest	relative	0.387	0.764	0.755
W or F	gru	absolute	0.387	0.735	0.729
W or F	gru	relative	0.387	0.765	0.763

S4 Intersecting groups

Table S4 gives the false-positive rate of protocol (B) for gender crossed with disability and with the most and least deprived IMD quintiles. With absolute features, disadvantages add up (women in the most deprived quintile: 0.496; men in the least deprived: 0.169). With relative features, the remaining gaps follow deprivation rather than gender (women and men in Q1: 0.260 and 0.234; men in Q5: 0.129).

Table S4: False-positive rates of protocol (B) for intersecting groups (25% cut-off), with the number of students not at risk in each group and 95% cluster-bootstrap intervals.

Features	Crossing	Group	n (negatives)	FPR [95% CI]
absolute	gender x disability	female × no disability	6,460	0.397 [0.386, 0.409]
absolute	gender x disability	female × disability	668	0.490 [0.450, 0.527]
absolute	gender x disability	male × no disability	7,716	0.205 [0.195, 0.214]
absolute	gender x disability	male × disability	538	0.297 [0.258, 0.335]
absolute	gender x IMD	female × Q1	1,327	0.496 [0.471, 0.525]
absolute	gender x IMD	female × Q5	1,223	0.346 [0.320, 0.373]
absolute	gender x IMD	male × Q1	1,194	0.290 [0.265, 0.315]
absolute	gender x IMD	male × Q5	1,729	0.169 [0.152, 0.187]
relative	gender x disability	female × no disability	6,460	0.183 [0.174, 0.193]
relative	gender x disability	female × disability	668	0.231 [0.200, 0.263]
relative	gender x disability	male × no disability	7,716	0.168 [0.159, 0.176]
relative	gender x disability	male × disability	538	0.240 [0.207, 0.276]
relative	gender x IMD	female × Q1	1,327	0.260 [0.238, 0.282]
relative	gender x IMD	female × Q5	1,223	0.134 [0.115, 0.155]
relative	gender x IMD	male × Q1	1,194	0.234 [0.208, 0.259]
relative	gender x IMD	male × Q5	1,729	0.129 [0.114, 0.145]

S5 Context descriptors associated with transfer

Table S5 lists the associations between context descriptors and the three transfer quantities that survive Benjamini–Hochberg control.

Table S5: Context descriptors associated with the three transfer axes (Spearman's ρ, Benjamini–Hochberg $q < 0.10$ within level, 69 tests in total). Mod.: modules; Pres.: presentations, which are not independent (they share the module-level LOMO model). Difficulty: AUC (A); Rank. cost: AUC (A) − AUC (B); Drift: flag-rate drift. Exploratory associations.

Level (n)	Axis	Descriptor	ρ	q
Mod. (7)	Difficulty	Share aged 35+	−0.96	0.016
Pres. (22)	Difficulty	Share aged 35+	−0.79	0.000
Pres. (22)	Difficulty	VLE activity types	+0.54	0.037
Pres. (22)	Difficulty	At-risk rate	+0.49	0.060
Pres. (22)	Rank. cost	Assessments due before cut-off	+0.57	0.030
Pres. (22)	Rank. cost	First deadline	−0.45	0.097
Pres. (22)	Drift	Withdrawn share among at-risk	−0.68	0.008
Pres. (22)	Drift	Share aged 35+	+0.64	0.015
Pres. (22)	Drift	At-risk rate	−0.61	0.023
Pres. (22)	Drift	Median clicks	−0.59	0.028
Pres. (22)	Drift	Share without A-level	+0.56	0.033
Pres. (22)	Drift	Assessment-structure distance	+0.52	0.048
Pres. (22)	Drift	VLE activity types	−0.51	0.048
Pres. (22)	Drift	Assessments due before cut-off	−0.44	0.099

Table S6: Model families and temporal information at the 25% cut-off (students still enrolled; means over modules). Forest and Boosting: aggregated features; Boosting + seq.: aggregated features plus weekly clicks and active days over the complete weeks before the cut-off and trend features; GRU: recurrent network on the weekly sequence plus all static features. Cost: AUC (A) − AUC (B); Drift: mean absolute flag-rate drift; GGG: drift in module GGG.

Features	Model	AUC (A)	AUC (B)	Cost	Drift	GGG
Absolute	Forest	0.800	0.787	+0.014	0.135	+0.51
	Boosting	0.805	0.787	+0.018	0.137	+0.53
	Boosting + seq.	0.808	0.791	+0.017	0.131	+0.52
	GRU	0.799	0.792	+0.007	0.127	+0.47
Relative	Forest	0.816	0.808	+0.008	0.021	+0.00
	Boosting	0.818	0.811	+0.007	0.051	+0.01
	Boosting + seq.	0.819	0.812	+0.007	0.054	+0.02
	GRU	0.817	0.815	+0.003	0.015	+0.00

S6 Other learners and sequences

Gradient boosting and a GRU network on weekly activity sequences reproduce the drift with absolute features and its correction with relative features (Table S6, Fig. S1). Temporal information adds little to early prediction: on unseen modules, the GRU exceeds the forest by 0.006 to 0.008 in AUC across cut-offs.

S7 Target definition

Restricted to students still enrolled at the cut-off, "Withdrawn only" becomes future withdrawal, a rarer and harder target (Table S7). The pattern holds: relative features lower the ranking cost from 0.028 to 0.012 and the mean absolute drift from 0.054 to 0.028, and raise the AUC on unseen modules from 0.623 to 0.662.

S8 Further robustness checks

Threshold. With absolute features, the mean absolute drift stays between 0.084 and 0.138 for every threshold from 0.3 to 0.7, and the drift of GGG between +0.16 and +0.60; with relative features, between 0.017 and 0.032.

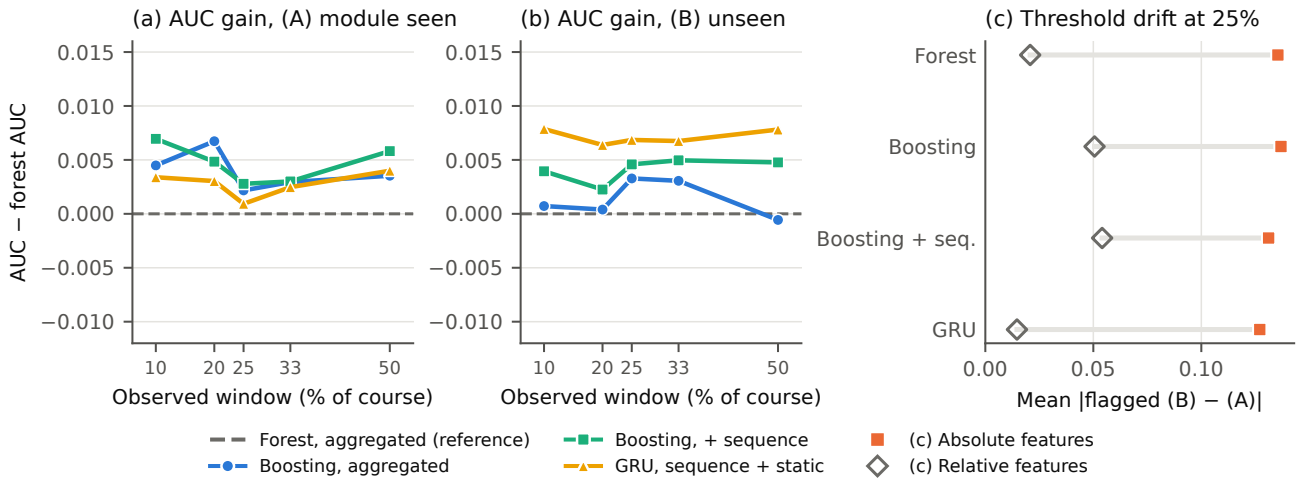


Figure S1: Sequences and other learners. (a, b) AUC gain over the aggregated random forest (dashed zero line) with relative features, when the module is seen (A) or unseen (B). (c) Mean absolute flag-rate drift at the 25% cut-off with absolute and relative features.

Table S7: Robustness to the target definition (25% cut-off, students still enrolled; means over modules). W or F: Withdrawn or Fail; W only: Withdrawn only. Abs./rel.: absolute or relative features. Cost: AUC (A) − AUC (B); Drift: mean absolute flag-rate drift.

Target	Feat.	Risk (%)	AUC (A)	AUC (B)	Cost	Drift	GGG
W or F	abs.	38.7	0.800	0.787	+0.014	0.135	+0.51
W or F	rel.	38.7	0.816	0.808	+0.008	0.021	+0.00
W only	abs.	13.7	0.651	0.623	+0.028	0.054	+0.17
W only	rel.	13.7	0.675	0.662	+0.012	0.028	+0.03

Seeds. Over 10 seeds (forest and fold assignment), the standard deviation of a module's AUC on unseen modules is at most 0.004 and that of its drift at most 0.010.

Standard versus strict hold-out. Keeping in the training modules the students also enrolled in the held-out module changes the mean AUC on unseen modules by at most 0.003. For the modules that share many students, removing them changes the AUC of CCC by −0.007 and of EEE by −0.006, whereas removing as many random enrolments changes it by +0.001 and −0.001.

Training size. The training set of (B) is 0.93 to 1.22 times that of (A). When the model that has seen the module is trained on exactly as many students as (B), the transfer cost is 0.014 with absolute features and 0.008 with relative features, as in the main analysis.